

Principles of Information Security

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Signature Schemes

- A *signature scheme* is a five-tuple $(\mathcal{P}, \mathcal{A}, \mathcal{K}, \mathcal{S}, \mathcal{V})$, where the following conditions are satisfied:
 - 1. $\mathcal{P}$ is a finite set of possible messages;
 - 2. $\mathcal{A}$ is a finite set of possible signatures;
 - 3. $\mathcal{K}$, the keyspace, is a finite set of possible keys;
 - 4. For each $k \in \mathcal{K}$, there is a signing algorithm $sig_k \in \mathcal{S}$ and a corresponding verification algorithm $ver_k \in \mathcal{V}$. Each $sig_k : \mathcal{P} \rightarrow \mathcal{A}$ and $ver_k : \mathcal{P} \times \mathcal{A} \rightarrow \{true, false\}$ are functions such that the following equation is satisfied for every message $x \in \mathcal{P}$ and for every signature $y \in \mathcal{A}$:
$$ver_k(x, y) = true, \text{ if } y = sig_k(x),$$
$$ver_k(x, y) = false, \text{ if } y \neq sig_k(x).$$
- The pair (x, y) with $x \in \mathcal{P}$ and $y \in \mathcal{A}$ is called a *signed message*.

The Digital Signature Algorithm (DSA)

- The DSA is based on the difficulty of computing logarithms and is based on schemes originally presented by ElGamal and Schnorr.

Table: Global Public-Key Components

p	prime number where $2^{L-1} < p < 2^L$ for $512 \leq L \leq 1024$ and L is a multiple of 64.
q	prime divisor of $(p - 1)$, where $2^{159} < q < 2^{160}$; i.e., bit length of 160 bits.
g	$= h^{(p-1)/q} \bmod p$, where h is any integer with $1 < h < p - 1$ such that $h^{(p-1)/q} \bmod p > 1$.

The Digital Signature Algorithm (DSA) (Continued...)

Table: User's Private Key

x	random or psuedo-random integer with $0 < x < q$.
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Table: User's Public Key

$y = g^x \bmod p.$

Table: User's Per-Message Secret Number

k	random or psuedo-random integer with $0 < k < q$.
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The Digital Signature Algorithm (DSA) (Continued...)

Table: Signing Phase

$$\begin{aligned}r &= (g^k \bmod p) \bmod q \\s &= [k^{-1}(H(M) + x.r)] \bmod q \\ \text{Signature} &= (r, s) \\ \text{Send } (M, (r, s)) &\text{ to reviewer.}\end{aligned}$$

M : message to be signed

Digital Signatures

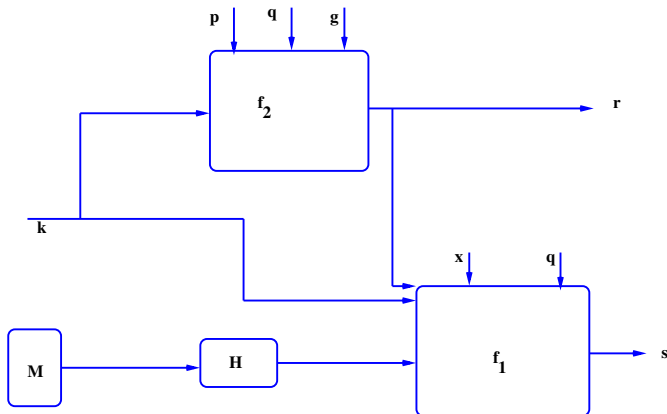


Figure: (a) Signing

$$s = f_1(H(M), k, x, r, q) = [k^{-1}(H(M) + x.r)] \bmod q$$
$$r = f_2(k, p, q, g) = (g^k \bmod p) \bmod q$$

The Digital Signature Algorithm (DSA) (Continued...)

Table: Verification Phase

$$w = (s')^{-1} \bmod q$$

$$u1 = [H(M').w] \bmod q$$

$$u2 = (r').w \bmod q$$

$$v = (g^{u1}.y^{u2} \bmod p) \bmod q$$

TEST: $v = r'$. If so accept; otherwise reject.

M', r', s' : received versions of M, r, s

Digital Signatures

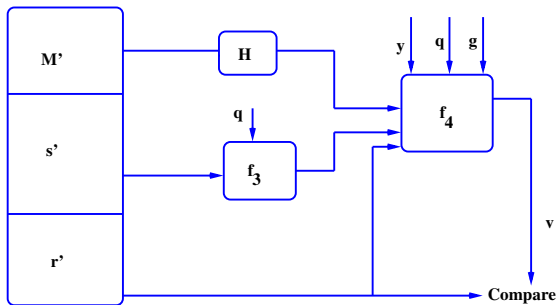


Figure: (b) Verifying

$$\begin{aligned} w &= f_3(s', q) = (s')^{-1} \bmod q \\ v &= f_4(y, q, g, H(M'), w, r') \\ &= ((g^{(H(M') \cdot w) \bmod p}) \cdot y^{(r' \cdot w) \bmod q}) \bmod p \bmod q \end{aligned}$$

The RSA Digital Signature

- The The RSA Digital Signature algorithm is based on the difficulty of factoring a composite number into two prime factors.

Table: Key generation of the RSA digital signature algorithm

Select p, q	p and q both prime, $p \neq q$ (p and q are large)
Calculate $n = p \times q$	
Calculate $\phi(n) = (p - 1)(q - 1)$	
Select integer e	$\gcd(e, \phi(n)) = 1; 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \pmod{\phi(n)}$
Public key	$KU = \{e, n\}$
Private key	$KR = \{d, n\}$

The RSA Digital Signature (Continued...)

Setup Phase

- $n = p \times q$
- $\mathcal{P} = \mathcal{A} = \mathcal{Z}_n$
- Define the key space as
$$\mathcal{K} = \{(n, p, q, e, d) : n = pq; p, q \text{ primes, } ed \equiv 1 \pmod{\phi(n)}, \gcd(e, \phi(n)) = 1\}$$
- Let two users A and B be Alice and Bob, respectively.
- Alice publishes $\{e, n\}$ and keeps $\{d, n\}$ as private.
- $x \in \mathcal{P}$ be a message to be signed by the Alice (User A).

The RSA Digital Signature (Continued...)

Signature Generation Phase

- A computes $y = \text{sig}_K(x) = x^d \pmod{n}$.
- The signature on the message x is y .
- The signed message on x is (x, y) .
- $Alice \rightarrow Bob : (x, y)$.

The RSA Digital Signature (Continued...)

Signature Verification Phase

- B computes $x' = y^e \pmod n$.
- Bob accepts signature as valid if $x' = x$.
- Bob otherwise rejects the signature.

The RSA Digital Signature (Continued...)

Correctness Proof of Signature Verification Phase

$$\begin{aligned}x' &= y^e \pmod n \\&= (x^d \pmod n)^e \pmod n \\&= x^{ed} \pmod n, \text{ since } ed \equiv 1 \pmod{\phi(n)} \\&= x^{k\phi(n)+1} \pmod n \\&= x.\end{aligned}$$

The ElGamal Digital Signature Scheme

Setup Phase

- p is selected as a large prime so that the discrete logarithm problem (DLP) in Z_p is intractable.
- $\alpha \in Z_p^*$ is a primitive root of p .
- $\mathcal{P} = Z_p^*$.
- $\mathcal{A} = Z_p^* \times Z_{p-1}$.
- Key space $\mathcal{K} = \{(p, \alpha, a, \beta) : 1 < a < p, \beta = \alpha^a \pmod{p}\}$.
- The public key is (p, α, β) .
- The private key is a .

The ElGamal Digital Signature Scheme (Continued...)

Signature Generation Phase

- Let $x \in Z_p^*$ be a message to be signed by the signer.
- Signer A selects a random number $k \in Z_{p-1}^*$.
- Signer A computes $\gamma = \alpha^k \pmod{p}$.
- Signer A computes $\delta = (x - a\gamma)k^{-1} \pmod{p-1}$.
- Signature on message x is $sig_K(x) = (\gamma, \delta)$.
- Signer sends the signed message $(x, (\gamma, \delta))$ to the verifier V .

The ElGamal Digital Signature Scheme (Continued...)

Signature Verification Phase

- Verifier V has $x, \gamma \in \mathbb{Z}_p^*, \delta \in \mathbb{Z}_{p-1}$, and public key (p, α, β) .
- Verifier V computes $\beta^\gamma \pmod{p}$.
- Verifier V computes $\gamma^\delta \pmod{p}$.
- Verifier V checks the condition: $\beta^\gamma \gamma^\delta \equiv \alpha^x \pmod{p}$.
- If above holds, V accepts A 's signature; otherwise, rejects it.

The ElGamal Digital Signature Scheme (Continued...)

Correctness Proof of Signature Verification Phase

- We have,

$$\delta = (x - a\gamma)k^{-1} \pmod{p-1}$$

$$k\delta = x - a\gamma \pmod{p-1}$$

$$a\gamma + k\delta = x \pmod{p-1}$$

- Now,

$$\begin{aligned}\beta^\gamma \gamma^\delta &= (\alpha^a \pmod{p})^\gamma (\alpha^k \pmod{p})^\delta \pmod{p} \\ &= \alpha^{a\gamma + k\delta} \pmod{p} \\ &= \alpha^x \pmod{p}\end{aligned}$$

The ElGamal Digital Signature Scheme (Continued...)

Example [ElGamal Digital Signature]

- Take a prime $p = 467$.
- Take a primitive root $\alpha = 2$.
- Choose randomly $a = 127$.
- Compute by repeated square-and-multiply algorithm,

$$\begin{aligned}\beta &= \alpha^a \pmod{p} \\ &= 2^{127} \pmod{467} \\ &= 132.\end{aligned}$$

- The public key is $(p, \alpha, \beta) = (467, 2, 132)$.
- The private key is $a = 127$.

The ElGamal Digital Signature Scheme (Continued...)

Example [ElGamal Digital Signature] (Continued...)

- Suppose Alice wants to sign the message $x = 100$.
- Alice chooses the random number $k = 213 \in \mathbb{Z}_p^*$. Note that $\gcd(k, p - 1) = \gcd(213, 466) = 1$.
- Alice computes $k^{-1} \pmod{p - 1} = 213^{-1} \pmod{466} = 431$, using the extended Euclid's GCD algorithm.
- Alice computes by repeated square-and-multiply algorithm,

$$\begin{aligned}\gamma &= \alpha^k \pmod{p} \\ &= 2^{213} \pmod{467} \\ &= 29,\end{aligned}$$

$$\begin{aligned}\delta &= (x - a\gamma)k^{-1} \pmod{p - 1} \\ &= (100 - 127 \times 29) \times 431 \pmod{466} \\ &= 51.\end{aligned}$$

The ElGamal Digital Signature Scheme (Continued...)

Example [ElGamal Digital Signature] (Continued...)

- Alice sends the signed message $(m, (\gamma, \delta)) = (100, (29, 51))$ to verifier, Bob.
- Bob verifies this signature by checking that $\beta^\gamma \gamma^\delta \equiv \alpha^x \pmod{p}$.
- Bob computes by repeated square-and-multiply algorithm,

$$\begin{aligned}\beta^\gamma \gamma^\delta &= 132^{29} \times 29^{51} \pmod{467} \\ &= 189, \\ \alpha^x &= 2^{100} \pmod{467} \\ &= 189.\end{aligned}$$

- Hence, the signature is verified.

Thank You